EEG-Based Intent-Driven Shared-Control Prosthetic Arm

Complete Teammate Integration Guide: Interface → ML → STM32 → Actuators

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1. System Architecture Overview

This project integrates real-time EEG decoding with an embedded robotic prosthetic arm. The Python interface runs the signal processing, feature extraction, and machine learning pipeline, then transmits structured serial commands to an STM32/Arduino microcontroller to actuate the hand.

```
[PhysioNet EEG 160Hz] → [CAR + 8-30Hz Filter] → [5-Subband FBCSP] → [Shrinkage LDA]
■
[Shared-Control Layer]
(Gate ≥ 75%, Debounce N=2, Lockout)
■
USB UART (<CMD:GRASP,CRC:0x4A>)
▼
[Prosthetic Arm (Servos/DC Motor)] ← [H-Bridge / PWM] ← [STM32 / Arduino Microcontroller]
```

2. Phase 1: Environment Setup on Your PC

Step 1.1: Extract interface_eeg.zip to any folder (e.g. C:\ProstheticArm\interface_eeg).

Step 1.2: Open PowerShell or Command Prompt in that directory and install requirements:

```
pip install -r requirements.txt
```

Dependencies: streamlit, plotly, pyserial, mne, scikit-learn, numpy, scipy, joblib.

3. Phase 2: Flashing Firmware to Microcontroller

Open firmware/prosthetic_arm_controller.ino in the **Arduino IDE**. Select your board (STM32 BluePill / Nucleo / Arduino Uno) and upload the code.

Important: Close the Arduino IDE Serial Monitor after uploading! If the Serial Monitor remains open, Windows locks the COM port, causing an 'Access Denied' error in Python.

4. Phase 3: Hardware Wiring & Pinout

Function	STM32 Pin	Arduino Pin	Connects To
Motor Close (PWM A)	PA0	Pin 5	H-Bridge IN1 / PWM
Motor Open (PWM B)	PA1	Pin 6	H-Bridge IN2
Servo Signal	PA8	Pin 9	Servo Orange/Yellow Wire
Heartbeat LED	PC13	Pin 13	Built-in LED (Blinks at 2Hz)
Common Ground	GND	GND	Driver GND & Battery GND

Power Isolation Rule: NEVER power high-torque motors directly from the microcontroller's 3.3V/5V rail! Use an external battery or bench supply. Connect the external battery GND to the microcontroller GND (Mandatory Common Ground).

5. Phase 4: Launching Interface & Stage A Verification

Step 4.1: Double-click run_interface.bat. The dashboard automatically launches at http://localhost:8501.

Step 4.2: Hardware Verification (Stage A):

1. In the sidebar under **Actuator Mode**, switch from *Simulation Mode* to **Physical Hardware (STM32/UART)**.
2. Choose your microcontroller's COM port from the dropdown.
3. Click **[Initialize Port]**. Status changes to CONNECTED (115200 baud).
4. Click **[Close Hand]**: Sends <CMD:GRASP,CRC:0x4A>\n. Hand physically closes.
5. Click **[Open Hand]**: Sends <CMD:RELEASE,CRC:0x5B>\n. Hand physically opens.

6. Phase 5: Autonomous EEG Intent Decoding (Stage B Verification)

Once physical motion is verified, start the real-time continuous neural decoding pipeline:

1. Click **[Start Stream]** in the top control bar.
2. The continuous EEG oscilloscope displays live 14-channel sensor readings with the shaded 2.0s epoch box and amber pointer (▼ POINTER (t)).
3. **FBCSP + LDA Pipeline:** Computes real-time intent probabilities for REST, GRASP, and RELEASE.
4. **Shared-Control Layer:** Requires confidence ≥ 75% across 2 consecutive windows ([OK] [OK] debounce) before triggering the physical arm.
5. **Quick Test Cues:** Click **[Jump to Grasp Event (@ 4.2s)]** to watch the classifier detect grasp intent and actuate the arm automatically.

7. Serial UART Communication Protocol

Command Packet Sent	Hardware Action	Microcontroller Response
<CMD:GRASP,CRC:0x4A>\n	Closes fingers / grip	<ACK:GRASP_ACCEPTED>
<CMD:RELEASE,CRC:0x5B>\n	Opens fingers / release	<ACK:RELEASE_ACCEPTED>
<CMD:HOLD,CRC:0x3F>\n	Maintains position (idle motors)	<ACK:HOLD_ACCEPTED>
<CMD:STATUS,CRC:0x12>\n	Telemetry request	<STATUS:STATE=...,ANGLE=...>

8. Troubleshooting & Common Pitfalls

Symptom	Root Cause	Solution
Access Denied on COM Port	Port locked by another program	Close Arduino Serial Monitor / PuTTY before initializing.
COM Port missing from dropdown	Missing USB-UART drivers	Install CH340, CP2102, or STM32 VCP driver.
Board resets when motors move	Inrush current voltage drop	Use separate motor battery; add 470uF-1000uF capacitor.
Arm stays in HOLD state	Mechanical lockout active	When already closed, grasp pulses are suppressed for motor safety.